



- Meningitis
- Encephalitis
- Brain abscess
- Cerebral toxoplasmosis (specially in HIV positive patients)
- **A** - Autoimmune: Sarcoidosis
 - Neuro sarcoidosis → Development of lack of control over ADH release → ↑ ADH → Euvolemic Hyponatremia
- **A** - Alveoli infections: Legionella pneumophila (presents with confusion, disorientation, agitation)
 - Legionella pneumophila causes SIADH → ↓ Sodium to <125 mEq/L → Seizures
- **D** - Drugs:
 - SSRIs
 - Haloperidol
 - Vincristine
 - Chlorpropamide: Sulfonylurea drug, withdrawn from market d/t S/E of SIADH
- **H** - Hypothyroidism, Hypocortisolism, Hydrocephalus, Head injury
 - T3/T4/Cortisol, tend to have **inhibitory control** over release of ADH
 - Hydrocephalus/ Head injury: causes ↑ ICT → **Reset Osmostat**

Other Causes of Euvolemic Hyponatremia

- Excess consumption of water for rehydration after marathon running
- Beer potomania
- Hypothyroidism
- Hypocortisolism

Hypervolemic Hyponatremia

- Excess Aldosterone- More stimulation of Epithelial sodium channel leading reabsorption of salt & water and loss of K^+ , H^+
- Causes:
 - CHF: ↓ Cardiac output → ↓ GFR → RAAS activation
 - AKI, CKD, ATN, HUS: ↓ GFR → RAAS activation
 - Chronic Liver Disease: Ascites (d/t less Albumin) → RAAS activation
 - Nephrotic syndrome: Ascites → RAAS activation
- Diuretics are used for management

HYPOTONIC HYPONATREMIA	HYPOVOLEMIC (↓ TBW, ↓ TBS)	EUVOLEMIC (↑TBW, NORMAL TBS)	HYPERVOLEMIC (↑↑ TBW, ↑ TBS)
Causes	• Diarrhea • CSWS • Excess Diuretics	• Small Cell Ca Lung • Cerebral Toxoplasmosis • ↓ T4 • Marathon runners	• CHF • CLD • CKD



- | | | |
|---|---|---|
| <ul style="list-style-type: none"> • Mild dehydration: ORS • Severe dehydration: RL • Vomiting: NS | <ul style="list-style-type: none"> • Fluid restriction • Vaptans | <ul style="list-style-type: none"> • Fluid restriction • Diuretics <ul style="list-style-type: none"> ○ CHF and CLD: Spironolactone ○ CKD: Metolazone / Chlorthalidone |
|---|---|---|

HYPERTONIC HYPONATREMIA & ISOTONIC HYPONATREMIA

00:25:15

HYPERTONIC HYPONATREMIA	ISOTONIC HYPONATREMIA / PSEUDOHYPONATREMIA
• Plasma Osmolality >295 mOsm/kg • Reason ○ Hyperosmolar coma: Very high sugar value Solvent drag → Sugar and sodium has an inverse relation → 100 mg/dL glucose ↑ = ↓ Na⁺ by 1.6 mEq/L ○ Mannitol (excess) ○ Radiocontrast agents	• Pseudohyponatremia can occur due to: ○ Hyperlipidemia ○ Paraproteinemia • Both interfere with sodium measurement, causing falsely low sodium levels

TREATMENT OF HYPONATREMIA

00:28:42

Acute onset Hyponatremia with seizures (Na ⁺ <125 mEq/L)	Chronic onset hyponatremia with seizures (Na ⁺ <125 mEq/L)
Case scenario: Patient ran first marathon of life. He drinks 5 L of water afterwards to cool down. In late evening he develops confusion, agitation and disorientation followed by convulsion episode. On admission: Na ⁺ 120 mEq/L and K ⁺ 3.0 mEq/L	Case scenario AIDS patients admitted with Cerebral toxoplasmosis for past 3 days continues to have seizure episodes. Na ⁺ 120 mEq/ L, K ⁺ 4.0 meq/L
• T/t: 3% saline (100 mL bolus)	• T/t: ○ 3% saline, slow correction over 48 hours ○ ↑Na⁺ in blood = 8 mEq/day

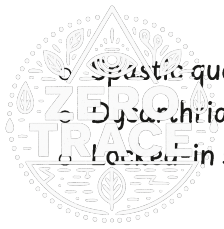
OSMOTIC DEMYELINATION SYNDROME (ODS)

00:31:54

- Formerly: Central Pontine Myelinolysis
- Cause: **Fast correction of chronic hyponatremia**

CLINICAL FEATURES

- 2-3 days after correction for hyponatremia was performed, patient develops
 - Confusion, behavioral changes



- Spastic quadriparesis (Corticospinal pathway damage)
- Dysarthria, dysphagia (Pons involvement)
- Locked-in syndrome (seen with Pontine damage)

WORKUP

- MRI Head: **Trident sign**

MCQ

Q. A 60 kg female patient has been sick for the past one week and was having multiple episodes of seizures. Electrolyte report shows Sodium = 120 mEq/L, Potassium = 4 mEq/L and Chloride = 90 mEq/L. Calculate the total sodium correction to be given to this patient?

- a. 20
- b. 200
- c. 400
- d. 600

Ans (d)

Solution:

- Formulae = $TBW \times [140 - \text{actual value}]$
 - $TBW \text{ (for male)} = Wt \times 0.6$
 - $TBW \text{ (for females)} = wt \times 0.5$
- For 60 Kg female patient, Total sodium correction = $(60 \times 0.5) \times [140 - 120] = 600 \text{ mEq}$
- 600 mEq slow correction over 48 hours

HYPERNATREMIA

00:36:52

- Value of **>158 mEq/L** can lead to seizures
- Seen M/c in elderly

CAUSES

- Physical disability: CVA, Alzheimer disease, Parkinsons disease, TKR | THR
- Diabetic insipidus:
 - Caused by both central and nephrogenic DI
 - DI causes Polyuria → Hypernatremia
- Excessive sweating
 - Disproportionate loss of water as compared to salt
- Lactulose excess
 - There will be osmotic diarrhoea
- Improperly diluted ORS
 - Concentrated ORS → Osmotic diarrhoea
 - Can present with "**Doughy Skin**"



ASYMPTOMATIC HYPERNATREMIA	SYMPTOMATIC HYPERNATREMIA (SEIZURES+)
Liberal water intake	- IV fluid: 5% Dextrose / 0.45% NS - Fluid should be given slowly over 48 hours Q. A 60 kg female is having seizures with $\text{Na}^+ = 160$ mEq/L. Calculate the total fluid to be given for correction: - Formula: $\text{TBW} \times [\text{actual value} - 140/140]$ - Calculation: $(60 \times 0.5) \times (160 - 140/140) = 30 \times (20/140) = 4.2 \text{ L}$

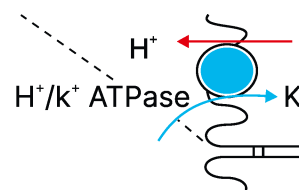
Important Information

- 5% Dextrose is isotonic initially but becomes hypotonic after glucose is metabolised inside the body
- Contraindicated in raised ICP
- IV fluid of choice for symptomatic hypernatremia
- Correction with IV fluids is done gradually

POTASSIUM

00:43:51

- Normal Values: 3.5–5.5 mEq/L
 - If <1.5 mEq/L → Diaphragmatic paralysis
 - If >8.8 mEq/L → Cardiac arrest
- In Diabetic Ketoacidosis (DKA)
 - Ketones → Stimulates vomiting centre → Hypokalemia d/t excessive vomiting
 - ↑ Sugar → Solvent drag
 - Insulin → Redistributes K^+ inside the cells
- Relation between K^+ and pH: **Inverse relation**
 - In hyperkalemia, $\text{H}^+ - \text{K}^+$ exchanger moves excess K^+ into cells, causing H^+ to move out into extracellular spaces → **acidosis**
 - If $\text{K}^+ \uparrow$ by 0.7 mEq/L, then the pH of blood $\downarrow$ by 0.1



HYPOKALEMIA

- Hypokalemia can lead to death due to:
 - Diaphragmatic paralysis
 - Torsades de Pointes | Polymorphic V-Tachycardia**
 - Broad RS complex
 - HR >100 bpm
 - Variable amplitude

ECG findings in Hypokalemia

- T wave inversion
- Prominent U wave





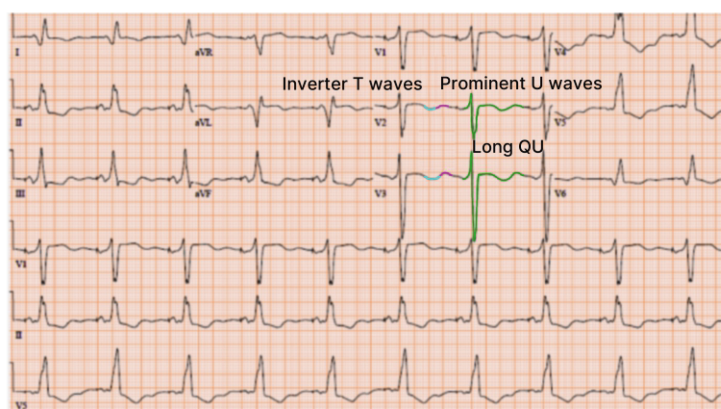
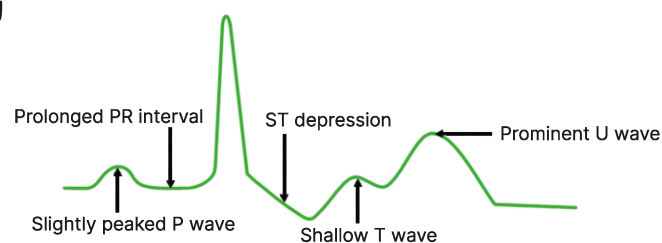
- Low potassium intake
- Alkalosis: Causes K^+ redistribution
- Sympathomimetic stimulation: Causes K^+ redistribution
 - Trauma
 - Salbutamol
 - Thyrotoxic periodic paralysis
- Renal wasting → **Kaliuria**
 - Bartter syndrome
 - Gitelman syndrome
 - Conn syndrome
 - B/L Adrenal hyperplasia
- Vomiting, diarrhea

CLINICAL FEATURES

- Muscle cramps & weakness
- Abdominal distention (Bowel sounds ↓)
- Flaccid weakness
- Paralytic ileus

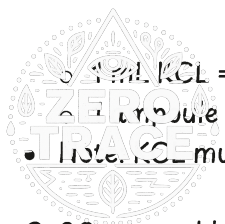
WORKUP:

1. ↓ Serum potassium
2. TTKG (Trans Tubular K^+ Gradient) >4 → Renal wasting
3. ECG changes
 - T wave inversion
 - Pseudo P-Pulmonale
 - Prominent U waves
 - ST depression



MANAGEMENT

- Correction for values between 3.0–3.5 mEq/L:
 - Oral correction → Oral Potchlor
- Correction for values < 3.0 mEq/L + reduced/ absent bowel sounds → **IV KCL**
 - 1 ampoule KCL = 10 mL



5 mL KCL = 2 mEq of KCL

1 ampoule containing 20 mEq will raise serum K^+ by ~ 0.25 mEq/L

- Note: KCL must be diluted before IV administration

Q. 30-year-old patient was started on Liposomal amphotericin B yesterday. On morning rounds you notice he is not able to sit up or lift arms. Serum electrolytes were ordered and serum K^+ 2.0 mEq/L. How much KCL in mEq should be given to this patient to achieve target value of 3.0 mEq/L?

- a. 20
- b. 40
- c. 80
- d. 160

Ans (c)

Explanation:

- 1 ampoule of KCl = 20 mEq
- 1 ampoule of KCl can raise serum K^+ by 0.25 mEq/L
- So, in order to raise serum K^+ values from 2.0 mEq/L to 3.0 mEq/L $\rightarrow$ 4 ampoules / 80 mEq of KCL need to be added to IV fluid

RULES TO BE FOLLOWED DURING POTASSIUM INFUSION

- Preferred fluid to be used for KCL infusion
 - KCL + NS is used
 - 5% dextrose is not used $\rightarrow$ Endogenous insulin release $\rightarrow$ Worsens pre-existing Hypokalemia
- Maximum KCL infusion via peripheral line: 10–15 mEq/L/hr
- Maximum KCL infusion via central line: 20 mEq/L/h
- In critical care setting 40 mEq can be used under intense monitoring

HYPERKALEMIA

01:02:19

- Cause of death if $K^+ > 8.0$ mEq/L: Diastolic arrest

CAUSES

- Pseudohyperkalemia:
 - Usage of thin bore needle, making a fist, $\uparrow$ TLC, $\uparrow$ RBC, $\uparrow$ Platelets can result in pseudohyperkalemia
- Hyperkalemic metabolic acidosis
 - AKI, CKD, ATN, HUS, RTA Type 4
- Drugs: ACE inhibitors, ARBs, Spironolactone
- Gordon syndrome:
 - Gain of function of Na^+Cl^- co-transporter $\rightarrow$ $\uparrow$ salt and water in body (HTN) $\rightarrow$ Negative feedback to RAAS $\rightarrow$ $\downarrow$ Aldosterone $\rightarrow$ Impaired excretion of K^+
 - Hypertension with hyperkalemic acidosis

EXTRA MILE

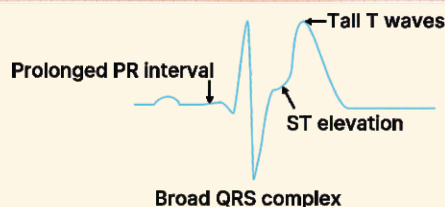
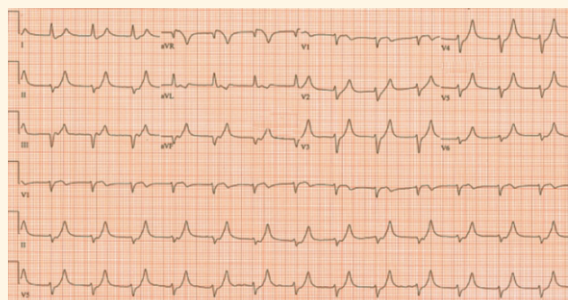
- Hypokalemic metabolic acidosis: Diarrhea



- **DOC** for Acute hyperkalemia: IV Calcium gluconate (antagonism), Calcium chloride

Important Information

- Characteristic feature of hyperkalemia on ECG
 - **Tall, tented T waves**: Earliest finding
 - ST elevation
 - Absent P-wave



- Drug causing maximum lowering of K^+ Levels: **Insulin drip** + 25-50% dextrose
 - Lower K^+ by **0.5-1 mEq/hour**
 - Dextrose causes endogenous release of Insulin
- Drugs causing loss of potassium in urine (Kaliuria):
 - Furosemide
 - Sodium zirconate
 - SPS (Sodium polystyrene sulfonate): **S/E of Colonic necrosis**
 - Patiromer
- Not routinely used in management: Soda bicarbonate (limited efficacy)

HYPOCALCAEMIA

01:11:02

- Normal value of Ca^{2+} : 9-11 mg%
- Tetany occurs if $Ca^{2+} < 7$ mg%
- Cause of death: **Laryngospasm**
- DOC for tetany: **IV 10% Calcium gluconate**
- Calcium chloride can also be used in the management of Hypocalcemia but Calcium carbonate is not used
- Clinical signs:
 - Trousseau sign
 - Chvostek sign



Carpopedal spasm
(Trousseau's sign)



Chvostek's sign

CAUSES

- Hypoparathyroidism
 - Leading cause: Autoimmunity
- Pseudohypoparathyroidism
 - GNAS gene → PTH receptor defect
- Inadvertent Parathyroid resection
- DiGeorge syndrome
 - Defective Thymus → ↓ in Cell mediated Immunity
 - Defect in development of Parathyroid gland → ↓ function of Parathyroid

HYPERCALCEMIA

- Leading cause of symptomatic hypercalcemia in hospitalized patients: **Malignancy**
 - **Squamous cell Lung cancer** → Production of PTH-rp → Causes bone resorption and thereby ↑ serum Ca^{2+}
 - Breast Cancer
 - Multiple myeloma
- Hypercalcemic crisis: $>15 \text{ mg\%}$
- Cause of death: Systolic arrest

STEPS IN MANAGEMENT OF HYPERCALCEMIC CRISIS

- 1st line management: NS + Furosemide (promote calciuria)
- **DOC** is Bisphosphonates: **Zoledronate**
 - Onset in 1-2 days and effect last for 3 weeks
 - Avoid if GFR <35
- Calcitonin nasal spray: **S/E Tachyphylaxis**
- Denosumab
- Oral phosphate
- Steroid responsive hypercalcemia
 - **Malignancy** (Multiple myeloma)
 - **Sarcoidosis**
 - Vitamin **D₃** intoxication
- Hemodialysis or peritoneal dialysis in case of renal failure

MAGNESIUM

01:18:00

- Normal Mg^{2+} : $1.3\text{--}2.1 \text{ mEq/L}$
- Death if <1 or $>10 \text{ mEq/L}$
- Mg^{2+} is **antagonist** to effects of intracellular calcium, so high magnesium will cause the following effect
 - Blood vessel relaxation
 - Airway relaxation
 - Reduce Neuromuscular transmission causing reflexes to become slower

HYPOMAGNESEMIA

CAUSES	• Alcoholic, Chronic diarrhea • Thiazides (Excess use) ◦ TRPM6 transporter in distal tubule is responsible for Mg^{2+} reabsorption ◦ Thiazides block TRPM6 → Mg^{2+} wasting • Drugs: Amphotericin B, Aminoglycosides • Gitelman syndrome: Defect of NCCT, TRPM6 transporter EXTRA MILE • Hypomagnesemia often coexists with hypokalemia, especially in malnourished children (PEM Grade 3 & 4)
CLINICAL FEATURES	• Muscle cramps • Tremors and nystagmus • HTN • Polymorphic VT risk is increased



- Serum $Mg^{2+} \downarrow$
- ECG: QT prolongation
- Oral Magnesium oxide / IV $MgSO_4$

HYPERMAGNESEMIA

01:23:05

CAUSES	• Chronic kidney disease • Eclampsia patients with $MgSO_4$ (Pritchard regimen) • Antacids and Laxative abuse
CLINICAL FEATURES	• Shock unresponsive to IV fluids • DTR $\downarrow$ • Urine output $\downarrow$ • Respiratory rate $\downarrow$
MANAGEMENT	• IV 10% Calcium gluconate

SUMMARY OF ELECTROLYTE IMBALANCES

01:24:46

CONDITION	CLINICAL MANIFESTATION	TREATMENT
$Na^+ < 125$ mEq/L	Seizures	3% saline
$Na^+ > 158$ mEq/L	Seizures	5% dextrose
$K^+ < 1.5$ mEq/L	Diaphragmatic Paralysis	KCl + NS infusion
$K^+ > 8.0$ mEq/L	Diastolic arrest	Calcium gluconate
$Ca^{2+} < 7$ mg/dL	Laryngospasm	Calcium gluconate
$Ca^{2+} > 15$ mg/dL	Systolic arrest	Bisphosphonates
$Mg^{2+} > 10$ mEq/L	Asystole	Calcium gluconate
$Mg^{2+} < 1$ mEq/L	Arrhythmia	IV $MgSO_4$



Q. Which of the following conditions is not associated with hyperkalemia?

- a. Gordon syndrome
- b. RBC lysis
- c. Liddle syndrome
- d. Addison's disease

Ans (c)

Q. A lung cancer patient is admitted with complaints of dizziness and hypotension. ECG shows sinus tachycardia with shortened QT interval. Which of the following is best for management?

- a. Bisphosphonates
- b. Pericardiocentesis
- c. Calcitonin nasal spray
- d. Furosemide drip

Ans (a)

Q. A 45-year-old woman after thyroid surgery on day 3 develops breathing difficulty and tingling sensation in her fingers and toes. On examination pulse 120/min with RR 20/min, BP 100/60 mmHg, and stridor is noted. The patient is at risk of developing which of the following?

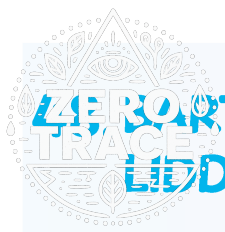
- a. Pulmonary thromboembolism
- b. Cardiac arrest
- c. Laryngeal edema
- d. Laryngospasm

Ans (d)

Q. Which of the following condition is the leading cause of hypernatremia?

- a. Hyperosmolar coma
- b. Diabetes insipidus
- c. Mannitol infusion
- d. Extreme debilitation

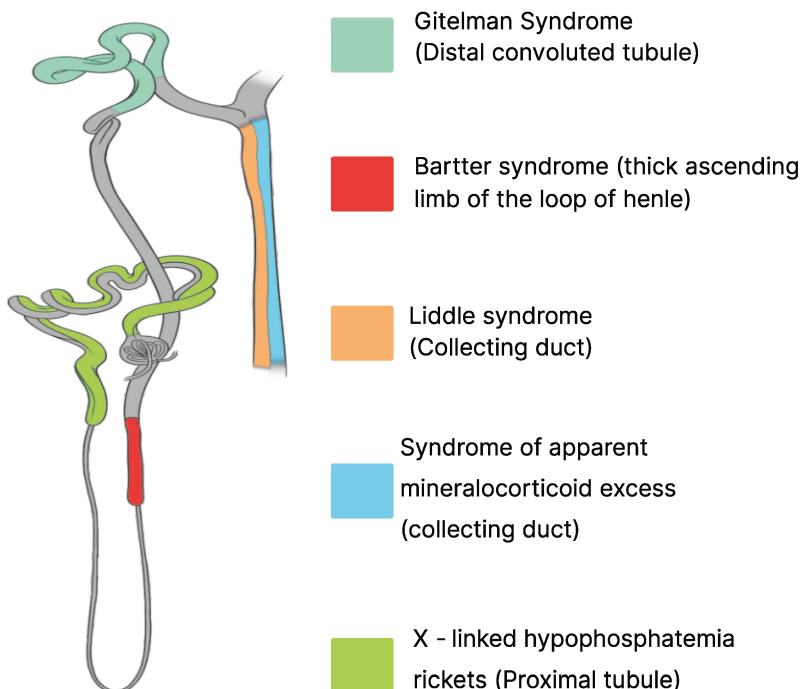
Ans (d)



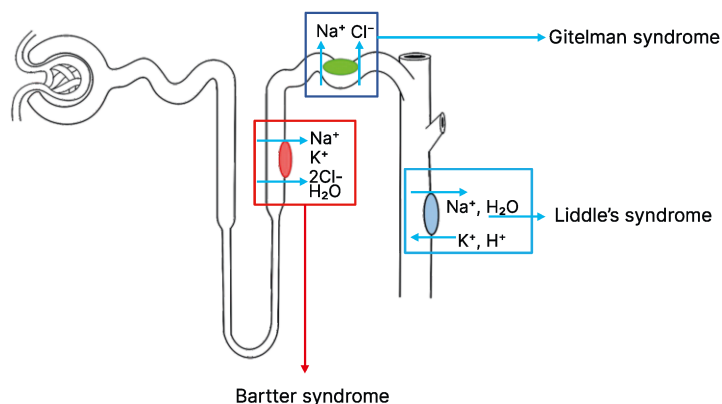
BARTTER SYNDROME, GITELMAN SYNDROME, LIDDLE SYNDROME, AND COMPARISON WITH SIADH

INTRODUCTION

00:00:40



GITELMAN SYNDROME	• Affects DCT • Loss of function of Na^+Cl^- cotransporter
GORDON SYNDROME	• Affects DCT • Gain of function of Na^+Cl^- cotransporter
BARTER SYNDROME	• Affects Thick ascending limb of LOH • Affects $\text{Na}^+\text{K}^+\text{Cl}^-$ cotransporter
LIDDLE SYNDROME	• Affects Collecting duct • Gain of function of Epithelial Na^+ channels • Manifestation: HTN, Hypokalemic alkalosis
SYNDROME OF APPARENT MINERALOCORTICOID EXCESS	• Affects Collecting duct • Cortisol stimulates the Epithelial Na^+ channels • Manifestation: HTN, Hypokalemic alkalosis
X-LINKED HYPOPHOSPHATEMIC RICKETS	• Affects Proximal tubule • Gene involved: PHEX gene • Severe manifestations of Rickets in pediatric age group



• Gitelman Syndrome

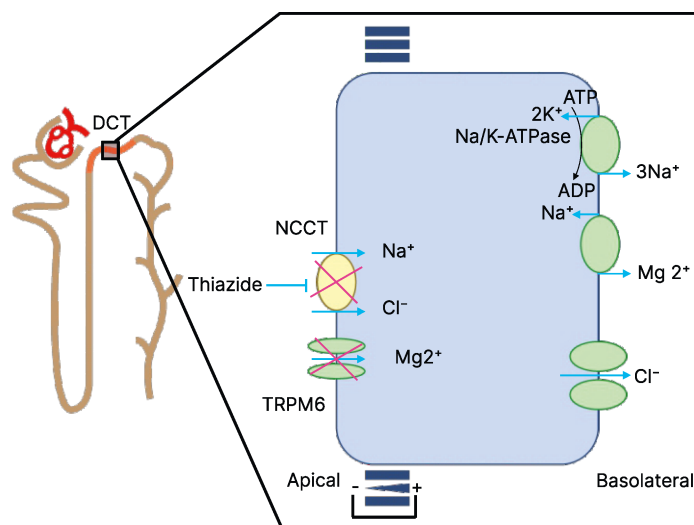
- Also affects TRPM6 transporter, responsible for Mg^{2+} reabsorption
 → Impaired TRPM6 function Mg^{2+} wasting → $\downarrow \text{PTH}$ → $\downarrow \text{Ca}^{2+}$ → **Tetany like manifestation**

GITELMAN SYNDROME

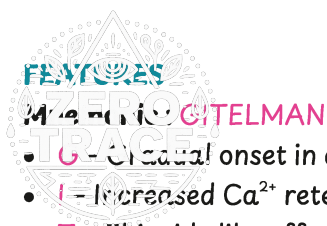
00:03:59

- Pattern of Inheritance : **Autosomal recessive**
- Gene: SLC12A3 inactivating mutations
- Manifestation mimics features of **Thiazide excess**

PATHOPHYSIOLOGY



- Defect in Na^+Cl^- cotransporter
 - Impaired Na^+ reabsorption → **Salt wasting, salt craving, polyuria**
 - Leads to **dehydration** → Activates **RAAS (2° hyperaldosteronism)**
 - Aldosterone stimulates **ENaC**, causing Na^+ reabsorption with **excess K^+ and H^+ loss** → **Hypokalemic alkalosis**
 - Impaired Cl^- reabsorption adds to **Hypochloremic hypokalemic alkalosis**
- Defect in TRPM6 transporter → Mg^{2+} wasting
 - Low Mg^{2+} → $\downarrow \text{PTH}$ secretion → Hypocalcaemia → Tetany like manifestation (Carpopedal spasm and Trousseau sign)
 - D/t $\downarrow \text{Ca}^{2+}$ in body → Increased Ca^{2+} retention → $\downarrow \text{Ca}^{2+}$ in urine → **Hypocalciuria**



- **G** - Gradual onset in adolescence with muscle weakness, cramps, and salt craving
- **I** - Increased Ca^{2+} retention: **Hypocalciuria**
- **T** - Thiazide like effects
- **E** - Electrolyte issue (Hypochloremic hypokalemic metabolic alkalosis) and fractional excretion of Cl^- is increased
- **L** - Low to normal BP
- **M** - $\text{Mg}^{2+} \downarrow$
- **A** - Aldosterone is increased but BP can be low
- **N** - Normal renal ultrasound, rules out use of diuretics

WORK UP

00:12:01

- Serum electrolytes: $\text{Na}^+ \downarrow$, $\text{K}^+ \downarrow$
- Urine osmolality: Low
- Spot sample urinary chloride: $\uparrow$
- Spot sample urine calcium to creatinine ratio: $\downarrow$
- Serum Mg^{2+} : $\downarrow$

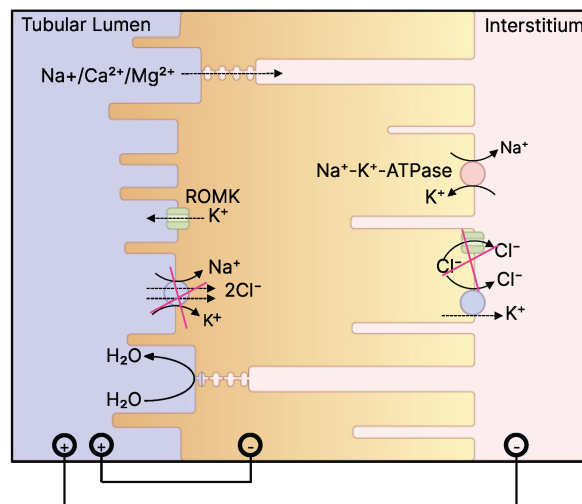
TREATMENT

- No definitive treatment
- Liberal salt intake, oral magnesium, and potassium supplements

BARTTER SYNDROME

00:14:01

- Antenatal form $\downarrow$ **Hyperprostaglandin E syndrome**
 - Features: Polyhydramnios and salt wasting
- Defect in Na^+Cl^- cotransporter and Cl^- channels on interstitial side
- Similar Cl^- channels in the inner ear maintain normal hearing; their involvement causes **SNHL**
- Loss of electrochemical gradient for calcium reabsorption via tubule is lost $\rightarrow$ **Hypercalciuria**
- Mimics Furosemide excess

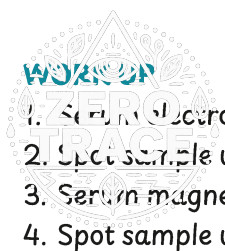


00:17:42

FEATURES

Buzz words: Child/infant with concomitant deafness presents with

- Failure to thrive
- Features of dehydration: Sunken eyeballs (polyuria $\rightarrow$ dehydration $\rightarrow$ activation of RAAS)
- Hypokalemic alkalosis
- BP low to normal
- Loss of electrochemical voltage gradient for calcium absorption leads to hypercalciuria
 - Features: Kidney stones, and Rickets like illness



1. Serum electrolytes: $\text{Na}^+ \downarrow$, $\text{K}^+ \downarrow$
2. Spot sample urinary chloride: $\uparrow$
3. Serum magnesium: Normal
4. Spot sample urinary calcium: $\uparrow$

TREATMENT

- No definitive treatment
- **Indomethacin** attenuates RAAS activation

COMPARISON OF GITELMAN SYNDROME WITH BARTER SYNDROME

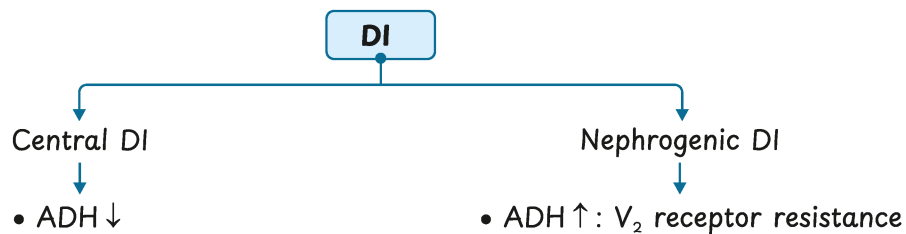
00:20:44

FEATURES	GITELMAN SYNDROME/ GENTLE	BARTER SYNDROME / BRUTAL
Onset	Childhood/Adult	Neonatal/Infantile
Affected segment	DCT	TAL
Calcium in urine	Hypocalciuria	Hypercalciuria
Electrolytes	Hypokalemic hypochloremic metabolic alkalosis	Hypokalemic hypochloremic metabolic alkalosis
Magnesium	Low	Normal
Main Symptoms	Cramps, Tetany like features and fatigue	Polyuria, dehydration, and growth delay

LIDDLE SYNDROME VS GORDON SYNDROME

00:22:29

	LIDDLE SYNDROME	GORDON SYNDROME
Genetics	• Autosomal dominant	• WNK-1 and WNK-4 genes are involved
Gain of Function of	• ENaC ◦ $\uparrow \text{Na}^+$ and water reabsorption ◦ $\uparrow \text{K}^+$, $\uparrow \text{H}^+$ excretion	• Na^+Cl^- cotransporter
BP	• Elevated	• Elevated ◦ $\uparrow \text{BP} \rightarrow$ Negative feedback to RAAS $\rightarrow \downarrow$ Aldosterone
Potassium	• Hypokalemia (Kaliuria)	• Hyperkalemia
Acidosis or alkalosis	• Metabolic Alkalosis	• Metabolic Acidosis

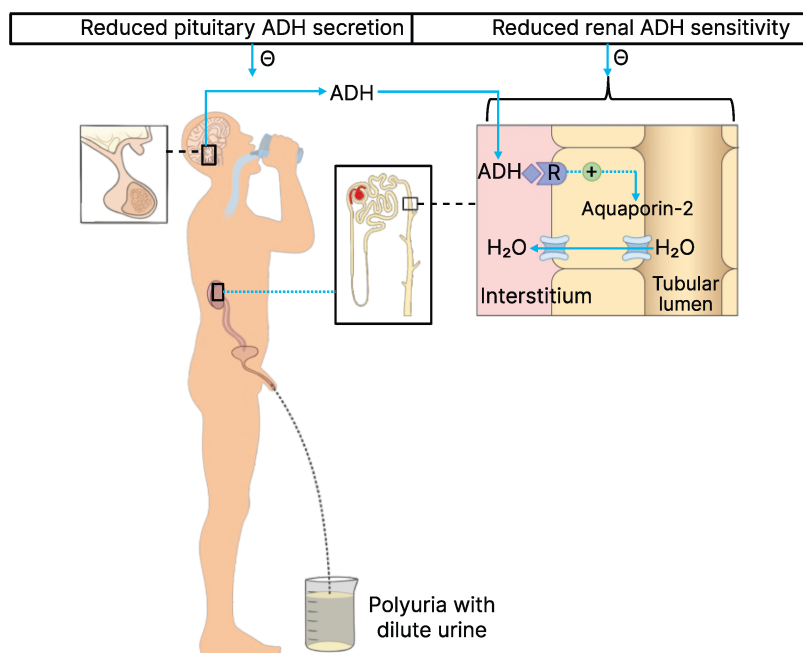


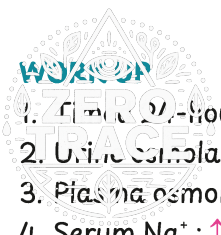
FEATURES

- Polyuria: Urine output >3 L/day
- Polydipsia: Water intake of >6 L/day
- Nocturia
- Dizziness and light-headedness: Loss of water + Orthostatic hypotension

CENTRAL DI	NEPHROGENIC DI
• Low ADH • Cause: ◦ Idiopathic ◦ Tumors: Causing damage to posterior pituitary ◦ Cranial surgery (post trans-sphenoidal surgery) → Transient DI ◦ Granulomas (Sarcoidosis): Sarcoidosis is a cause of both DI and SIADH ◦ Histiocytosis-X ◦ Arterio-venous malformation ◦ Sheehan syndrome	• V₂ receptor resistance • Causes: ◦ Lithium ◦ Hypokalemia: Hypokalemia of long-standing duration causes ↓ responsiveness of V₂ receptors to ADH ◦ Hypercalcemia: Calcium induced damage in CD lead to ↓ responsiveness of V₂ receptors to ADH ◦ Sickle cell anemia: ↓ blood supply to CD ◦ Amyloidosis

DI TYPES





1. 24-hour urine collection: Amount of urine $\uparrow\uparrow$
2. Urine osmolarity : $\downarrow$ (dilute)
3. Plasma osmolarity: $\uparrow$ (concentrated)
4. Serum Na^+ : $\uparrow$ (relative Hypernatremia)
5. Plasma ADH level
 - o Central DI : $\downarrow$
 - o Nephrogenic DI: $\uparrow$ / N
6. Water deprivation test (Miller-Moses test) : IOC

IN NORMAL INDIVIDUAL	IN DIABETES INSIPIDUS
• Miller-Moses test, if performed - o There will be 3% weight $\downarrow$ - o Urine becomes concentrated - o Urine osmolarity $\uparrow$ >2-4 times of plasma osmolarity	• Miller-Moses test, if performed - o All the finding seen in normal individual will not be seen - o Person will have dilute urine - o On administering ADH → If values normalize → Central DI → No Change → Nephrogenic DI

7. MRI T1 weighted: Loss of pituitary bright spot → Central DI



TREATMENT

00:38:37

- CDI: Desmopressin
- NDI:
 - o Thiazides: DOC
 - Rationale: They induce water volume depletion → Force PCT to absorb more water → Less water reaches CD and $\downarrow$ urine output (from >3L to <3L per day)
 - o Indomethacin
 - o Amiloride for Lithium induced Nephrogenic DI

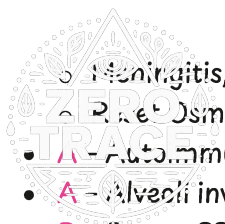
SIADH

00:41:59

CAUSES

Mnemonic : SIADH

- S - Small cell cancer of lung
- I - Infections (CNS):



- Meningitis, Encephalitis, Cerebral Toxoplasmosis
- **P** - **P**re-Osmostat: Total threshold ADH is lower in CNS infections
- **A** - **A**uto-immune: Sarcoidosis
- **A** - **A**lveoli involved in Pneumonia (*Legionella pneumophila*)
- **D** - **D**rugs: SSRI, CBZ, Vincristine, Chlorpropamide, Haloperidol
- **H** - **H**ypothyroidism (T3 and T4 have an inhibitory control on ADH release), Hydrocephalus, Head injury

DIAGNOSIS OF SIADH

00:46:22

ESSENTIAL CRITERIA FOR SIADH

- Plasma osmolality: <275 mOsm/kg H_2O
- **Clinical euvolemia**: No edema, Ascites, Orthostatic hypotension
- Urine osmolality: >100 mOsm/Kg H_2O
 - There is excess H_2O in the body $\rightarrow$ $\uparrow$ plasma volume $\rightarrow$ Negative feedback to RAAS $\rightarrow$ $\downarrow$ Aldosterone $\rightarrow$ Na^+ loss in urine
- Urine sodium >40 mmol/L on normal water and salt intake
- Normal KAT: KFT, Adrenal function test, Thyroid function test
 - Cortisol and Thyroid hormone has an inhibitory control on ADH release

SUPPLEMENTAL CRITERIA

00:51:34

- Inability to excrete 90% of 20 ml/Kg water load in 4 hours
- Serum vasopressin elevated in spite of low plasma osmolality
- Improvement of sodium with water restriction
- **$\downarrow$ BUN and $\downarrow$ Uric acid values**

IOC

- Water loading test

TREATMENT

- Water restriction.
- Vaptans: Tolvaptan

DIABETES INSIPIDUS VS SIADH

00:53:52

	SIADH	DIABETES INSIPIDUS	PSYCHOGENIC POLYDIPSIA	ADIPSIC HYPERNATREMIA
U. Osmolality	$\uparrow$	$\downarrow$ (dilute)	$\downarrow$	$\uparrow$
P. Osmolality (275-295 mOsm/kg)	$\downarrow$	$\uparrow$	$\downarrow$	$\uparrow$
Na^+ concentration	$\downarrow$	$\uparrow$	$\downarrow$	$\uparrow$



- Q. A 2-year-old child is brought with complaints of failure to gain weight. On examination he is having dehydration. Work up shows low sodium and hypokalemic metabolic alkalosis and increased urinary calcium. Mother also tells that child is deaf does not respond when she calls him by name. Diagnosis is?
- a. Gitelman syndrome
 - b. Bartter syndrome
 - c. Liddle syndrome
 - d. Gordon syndrome

Ans (b)



12. CILIOPATHIES



CILIA IN KIDNEY

FUNCTIONS

1. Chemosensors: Sensors for measuring the electrolyte traversing the tubule
2. Mechanosensors: Measures the volume of ultrafiltrate passing through a part of kidney

LOCATION

- Present in PCT, DCT, and Collecting duct
- Absent in the loop of Henle

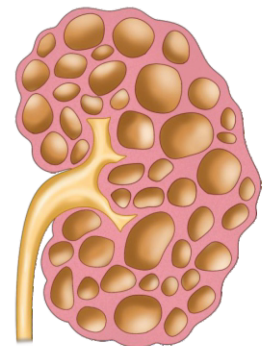
DISEASES RELATED TO KIDNEY CILIA

1. Autosomal Dominant Polycystic Kidney Disease (ADPKD)
2. Autosomal Recessive Polycystic Kidney Disease (ARPKD)
3. Medullary Sponge Kidney (MSK)
4. Autosomal Dominant Tubulointerstitial Kidney Disease
 - B/L shrunken kidneys
5. Nephronophthisis

IDENTIFICATION OF CILIOPATHIES IN PATHOLOGICAL SPECIMEN

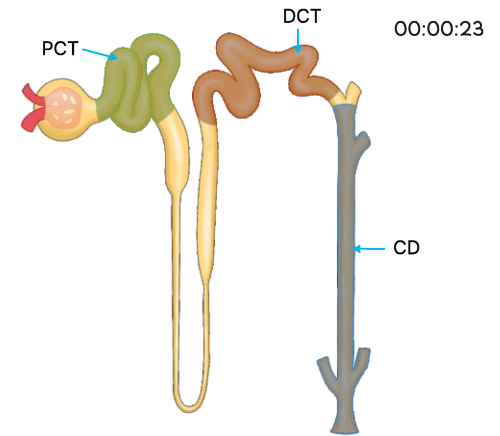
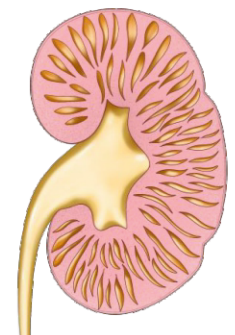
ADPKD

- Loss of corticomedullary differentiation; multiple cysts
- AD, B/L kidneys enlarged
- Cause of death: HTN, enhanced cardiovascular mortality
- Associated with **Hepatic cysts**



ARPKD

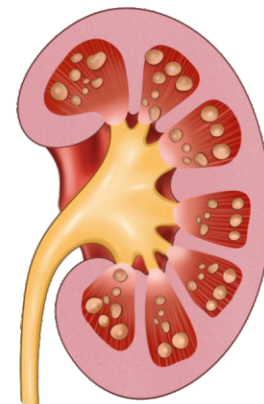
- Pediatric onset
- Radial striations on HPE
- Associated with **Hepatic fibrosis**
- B/L Enlarged Kidneys



00:01:16

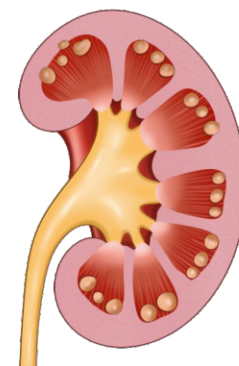


- Cysts in the medulla
- Swiss cheese pattern or bathing sponge pattern



ADTKD

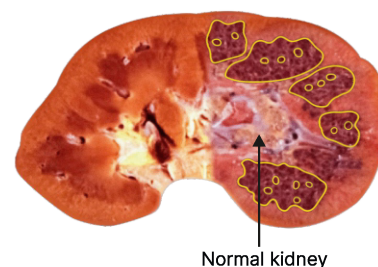
- Cysts at the corticomedullary junction
- Tubular atrophy present
- B/L shrunken kidney



MEDULLARY SPONGE KIDNEY (MSK)

- Sporadic in onset
- Cystic dilation of CD: Altered flow dynamics

00:05:57



CLINICAL FEATURES

- Age of presentation: 50-60 years

Mnemonic : SPONGE

- **S**: Stones in kidneys (calcium oxalate)
 - Cyst in medulla → Altered flow dynamics in CD → Stasis → Crystallization (Nephrolithiasis)
- **P**: Pyelonephritis → Fever, chills, **costovertebral tenderness**
 - On Urine culture: $>10^4$ CFU/mL + WBC cast
- **P**: Polyuria
 - D/t tubular damage, water will not be reabsorbed
- **O**: Blood in Urine
 - M/c cause of hematuria: Kidney stones
- **N**: Anemia: Low erythropoietin
 - D/t peritubular fibrosis
 - Normocytic normochromic anemia
- **G**: GFR is normal
- **E**: Erythropoietin is low